Audial Mind

SOC 2026

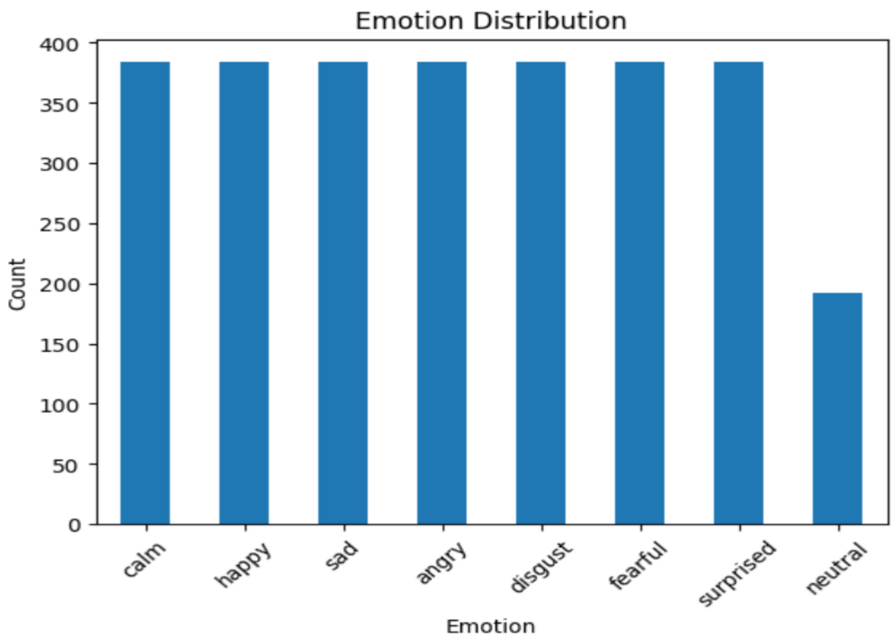
Name: Pulkit Gupta

Task 1

Part B

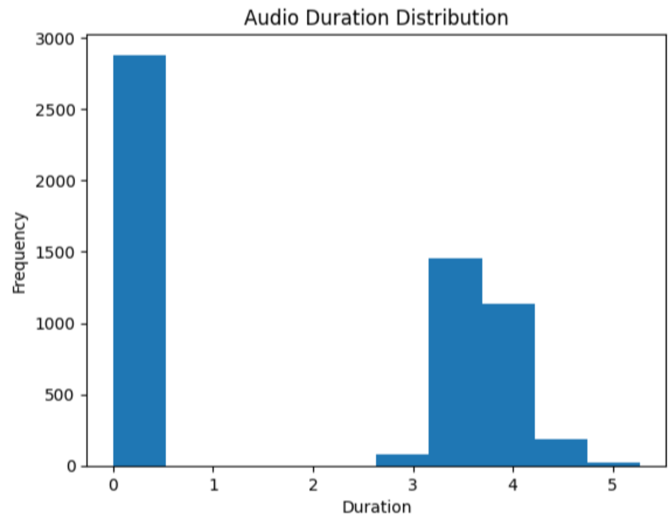
1. Files per emotion classes:

```
Emotion
calm          384
happy         384
sad           384
angry         384
disgust       384
fearful       384
surprised     384
neutral       192
```

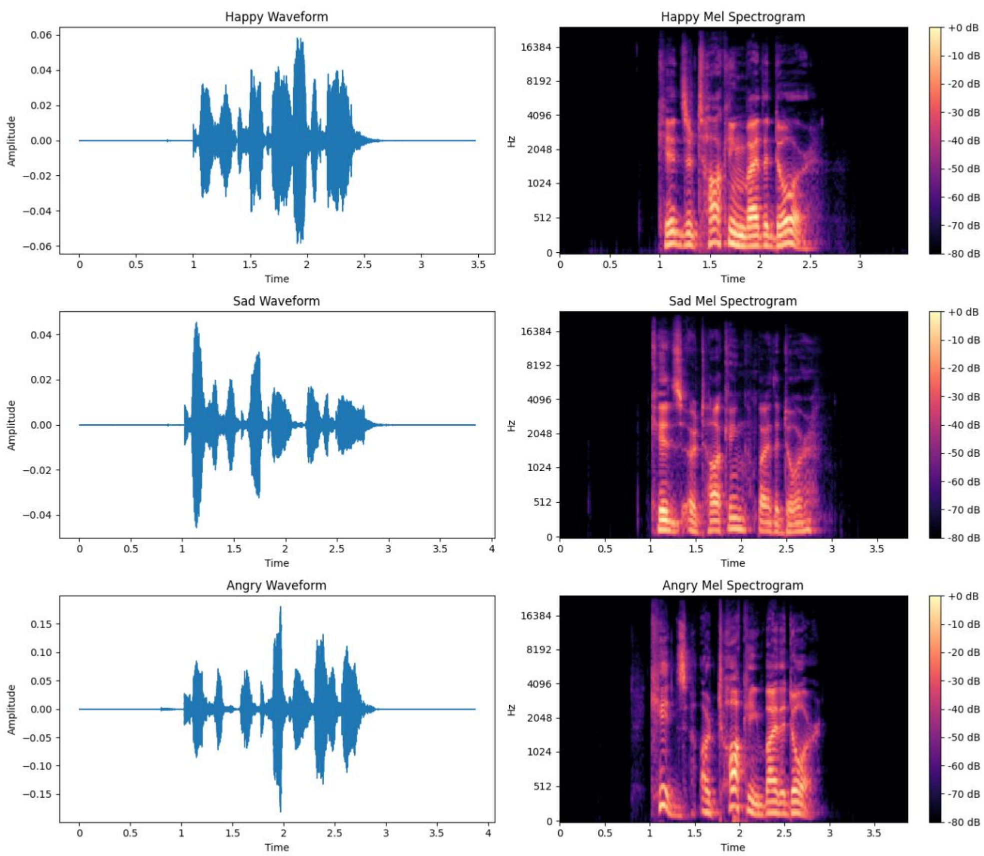


For all classes the distribution is same but there is less representation of neutral emotion in our data. This small imbalance can lead to model bias towards the majority classes, so some changes are needed.

2. Mean duration= 1.85 sec
   Std of duration= 1.86 sec



3. For happy, sad and angry:



If we compare the waveforms, we can see that amplitude of angry is highest showing louder speech, while happy has less amplitude while sad has the least amplitude showing soft speech. For the mel spectography, angry has more brighter at higher frequency which can be seen in waveform as well while brightness decreases in happy and then more in sad due to decrease in amplitude.

Part C

$UAR = \frac{1}{N}\sum_{i=1}^{N} Recall_i$

Recall$_i$ = $\frac{\mathrm{TP_i}}{TP_i+FN_i}$

Accuracy measures the performance of model over all classes regardless of the size, so if a class which is larger than any other class may dominate if we try to maximize the accuracy. Unweighted average recall calculates the recall for each class and since recall's numerator and denominator both depends on size of class, so for all class we get a value which are same in terms of magnitude and since this gives values of same magnitude for both big and small classes, it prevent form a bigger class in dominating the prediction of model. A problem of using normal accuracy can be seen by the example of where there are 2 classes of 95 (class A) and 5(class B) size, if the model predicts all 100 as class A, the normal accuracy will look 95% which looks good but we can see this is not a proper model.

Task 2

Part A

1. All 88 features

```
(1, 88)
['F0semitoneFrom27.5Hz_sma3nz_amean', 'F0semitoneFrom27.5Hz_sma3nz_stddevNorm', 'F0semitoneFrom27.5Hz_sma3nz_percentile20.0', 'F0semitoneFrom27.5Hz_sma
3nz_percentile50.0', 'F0semitoneFrom27.5Hz_sma3nz_percentile80.0', 'F0semitoneFrom27.5Hz_sma3nz_pctlrange0-2', 'F0semitoneFrom27.5Hz_sma3nz_meanRisingS
lope', 'F0semitoneFrom27.5Hz_sma3nz_stddevRisingSlope', 'F0semitoneFrom27.5Hz_sma3nz_meanFallingSlope', 'F0semitoneFrom27.5Hz_sma3nz_stddevFallingSlop
e', 'loudness_sma3_amean', 'loudness_sma3_stddevNorm', 'loudness_sma3_percentile20.0', 'loudness_sma3_percentile50.0', 'loudness_sma3_percentile80.0',
'loudness_sma3_pctlrange0-2', 'loudness_sma3_meanRisingSlope', 'loudness_sma3_stddevRisingSlope', 'loudness_sma3_meanFallingSlope', 'loudness_sma3_stdd
evFallingSlope', 'spectralFlux_sma3_amean', 'spectralFlux_sma3_stddevNorm', 'mfcc1_sma3_amean', 'mfcc1_sma3_stddevNorm', 'mfcc2_sma3_amean', 'mfcc2_sma
3_stddevNorm', 'mfcc3_sma3_amean', 'mfcc3_sma3_stddevNorm', 'mfcc4_sma3_amean', 'mfcc4_sma3_stddevNorm', 'jitterLocal_sma3nz_amean', 'jitterLocal_sma3n
z_stddevNorm', 'shimmerLocaldB_sma3nz_amean', 'shimmerLocaldB_sma3nz_stddevNorm', 'HNRdBACF_sma3nz_amean', 'HNRdBACF_sma3nz_stddevNorm', 'logRelF0-H1-H
2_sma3nz_amean', 'logRelF0-H1-H2_sma3nz_stddevNorm', 'logRelF0-H1-A3_sma3nz_amean', 'logRelF0-H1-A3_sma3nz_stddevNorm', 'F1frequency_sma3nz_amean', 'F1
frequency_sma3nz_stddevNorm', 'F1bandwidth_sma3nz_amean', 'F1bandwidth_sma3nz_stddevNorm', 'F1amplitudeLogRelF0_sma3nz_amean', 'F1amplitudeLogRelF0_sma
3nz_stddevNorm', 'F2frequency_sma3nz_amean', 'F2frequency_sma3nz_stddevNorm', 'F2bandwidth_sma3nz_amean', 'F2bandwidth_sma3nz_stddevNorm', 'F2amplitude
LogRelF0_sma3nz_amean', 'F2amplitudeLogRelF0_sma3nz_stddevNorm', 'F3frequency_sma3nz_amean', 'F3frequency_sma3nz_stddevNorm', 'F3bandwidth_sma3nz_amea
n', 'F3bandwidth_sma3nz_stddevNorm', 'F3amplitudeLogRelF0_sma3nz_amean', 'F3amplitudeLogRelF0_sma3nz_stddevNorm', 'alphaRatioV_sma3nz_amean', 'alphaRat
ioV_sma3nz_stddevNorm', 'hammarbergIndexV_sma3nz_amean', 'hammarbergIndexV_sma3nz_stddevNorm', 'slopeV0-500_sma3nz_amean', 'slopeV0-500_sma3nz_stddevNo
rm', 'slopeV500-1500_sma3nz_amean', 'slopeV500-1500_sma3nz_stddevNorm', 'spectralFluxV_sma3nz_amean', 'spectralFluxV_sma3nz_stddevNorm', 'mfcc1V_sma3nz
_amean', 'mfcc1V_sma3nz_stddevNorm', 'mfcc2V_sma3nz_amean', 'mfcc2V_sma3nz_stddevNorm', 'mfcc3V_sma3nz_amean', 'mfcc3V_sma3nz_stddevNorm', 'mfcc4V_sma3
nz_amean', 'mfcc4V_sma3nz_stddevNorm', 'alphaRatioUV_sma3nz_amean', 'hammarbergIndexUV_sma3nz_amean', 'slopeUV0-500_sma3nz_amean', 'slopeUV500-1500_sma
3nz_amean', 'spectralFluxUV_sma3nz_amean', 'loudnessPeaksPerSec', 'VoicedSegmentsPerSec', 'MeanVoicedSegmentLengthSec', 'StddevVoicedSegmentLengthSec',
'MeanUnvoicedSegmentLength', 'StddevUnvoicedSegmentLength', 'equivalentSoundLevel_dBp']
```

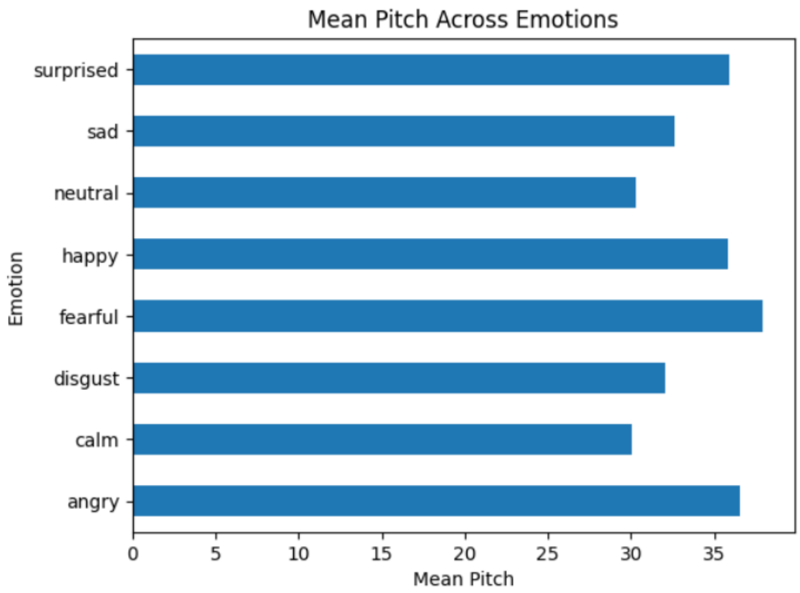
2. **F0semitoneFrom27.5Hz_sma3nz_amean** measures the average fundamental frequency (pitch) of the speech signal, expressed in semitones relative to 27.5 Hz.
   loudness_sma3_amean measures the average loudness or intensity of the speech signal over time.
   F2frequency_sma3nz_amean measures the average frequency of the second formant (F2), which is associated with tongue position during speech production.

Part B

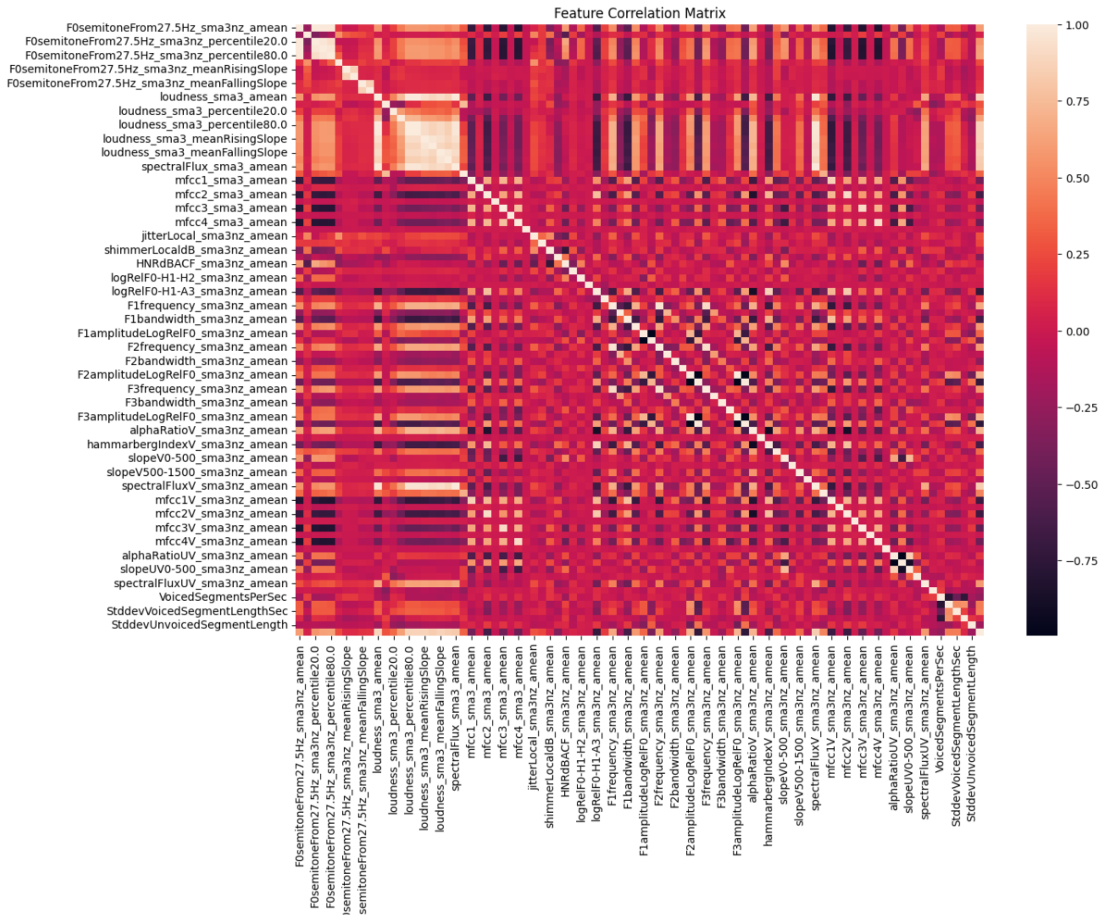
Shape of dataframe= (2880,89)

Part C

1. Fearful has the highest pitch closely followed by angry and surprised. These are emotions which are spoken with greater vocal tension and higher pitch so this is consistent with our intuition. Emotions like calm and neutral are expressed with relaxed speech so have less pitch.



2. The features with correlation greater than 0.9 are (F0semitoneFrom27.5Hz_sma3nz_amean <-> F0semitoneFrom27.5Hz_sma3nz_percentile20.0 = 0.977), (loudness_sma3_amean <-> loudness_sma3_percentile80.0 = 0.991)



Task 3

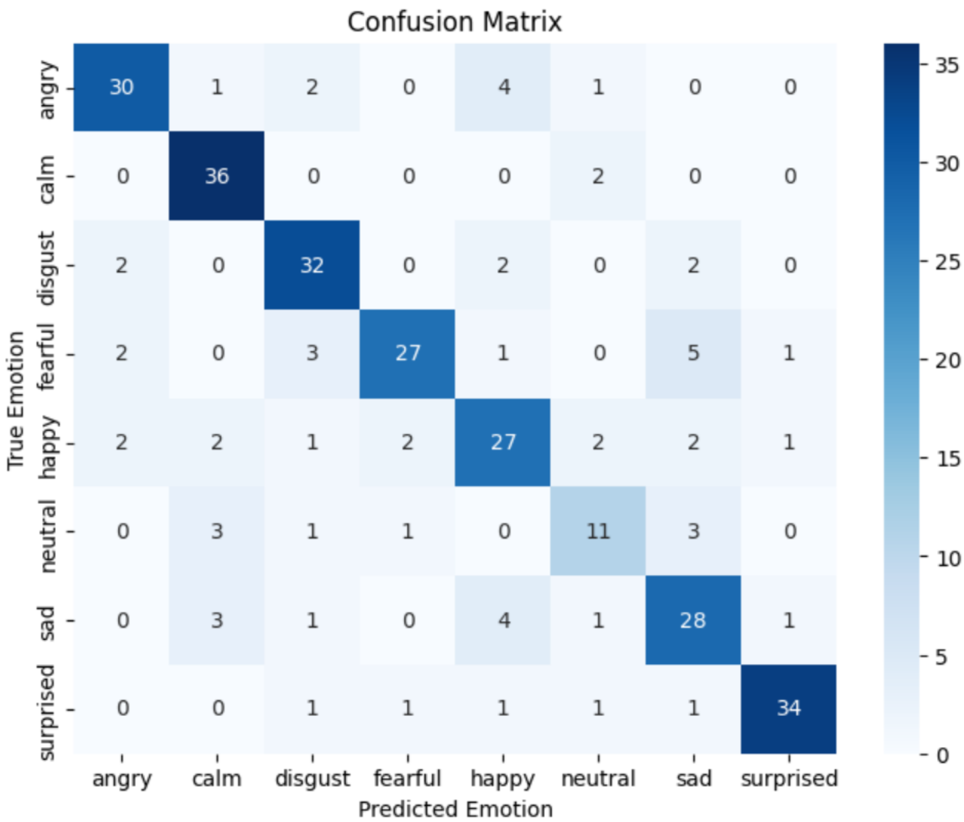
Part A

1. Using StandardScaler, we transform all the features to value between 0 and 1. This is important as SVM classifies based on distance and if we don't standardize, the features which are pretty large will have a lot more effect on model as compared to features with smaller value. Standardizing makes it almost equal in terms of magnitude so all features contribute equally in training and no feature dominates over other.

2.

```
Mean Accuracy: 0.7861111111111111
Accuracy Std: 0.015309310892394871
Mean UAR: 0.7731393387314439
UAR Std: 0.01471007266699816
```

3.



The most confused pair looks sad and fearful (5) followed by angry and happy, happy and sad (both 4). Most being fearful and sad makes sense because both are will low intensity, energy levels and pitch, angry and sad because both are with high energy level, pitch and speaking rate. Happy and sad doesn’t have same characterstics and are confused most likely due to different people have different ways of speaking.

4.

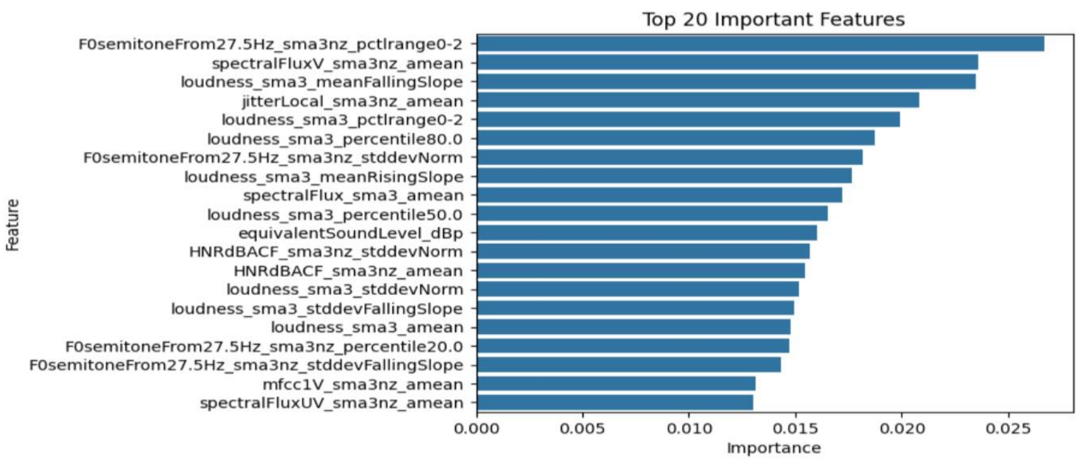
```
=====================
C = 0.1
======================
Mean Accuracy: 0.7632575757575757
Mean UAR: 0.7484081707765917

=====================
C = 1
======================
Mean Accuracy: 0.7647569444444443
Mean UAR: 0.7501152721547458

=====================
C = 10
======================
Mean Accuracy: 0.7799145299145298
Mean UAR: 0.7663137132772759
```

Part B

1. Mean UAR is a lot better than SVM

```
Mean Accuracy: 0.9638888888888888
Mean UAR: 0.9642695681511471
```

2.



3. Yes, the important features from our model are consistent with acoustic correlates of emotion. The main acoustic corelates of emotion are pitch, loudness and spectral features. These are the 3 most important features of our model, F0semitoneFrom27.5Hz_sma3nz_amean (for pitch), spectralFluxV_sma3nz_amean (for spectral features) and loudness_sma3_meanFallingSlope (for loudness). This features can be used to differentiate happy, surprised and anger with higher pitch and loudness to sad, calm and neutral for lower pitch and loudness. The presence of jitter and HNR features among the top predictors also suggests that voice quality contributes to emotion recognition.

Part C

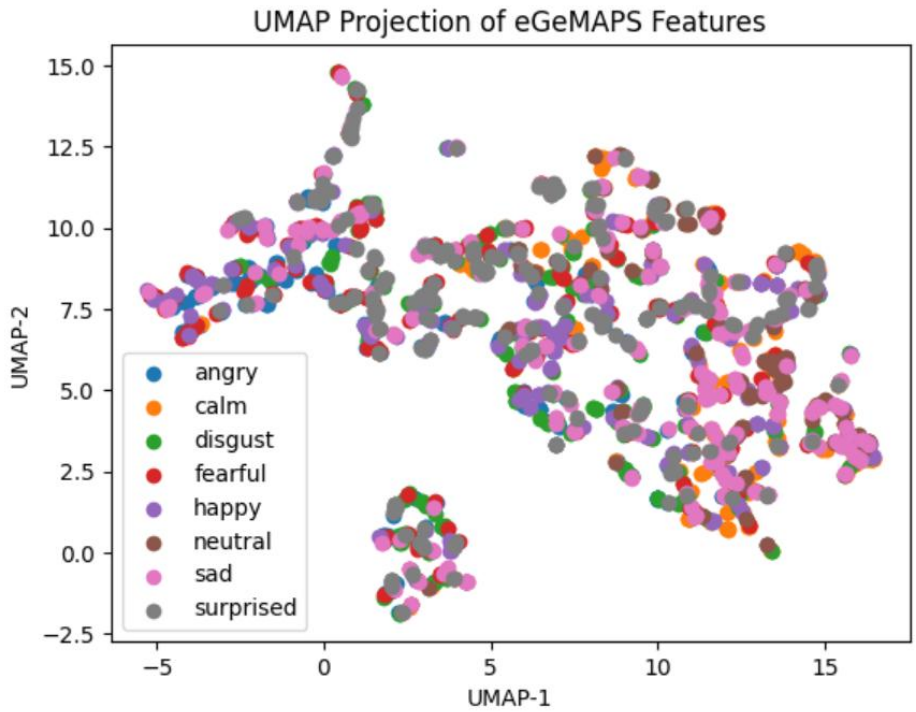
Model	Features	Mean Acc.	Mean UAR	UAR Std
SVM (RBF)	eGeMAPS	0.786	0.773	0.014
Random Forest	eGeMAPS	0.963	0.964	0.015

The model to be preferred for Audial Mind project should be random forest as its accuracy and UAR are a lot better as compared to SVM. The std being low also shows that the model have almost equal accuracy between different folds and not unstable results.

Task 4

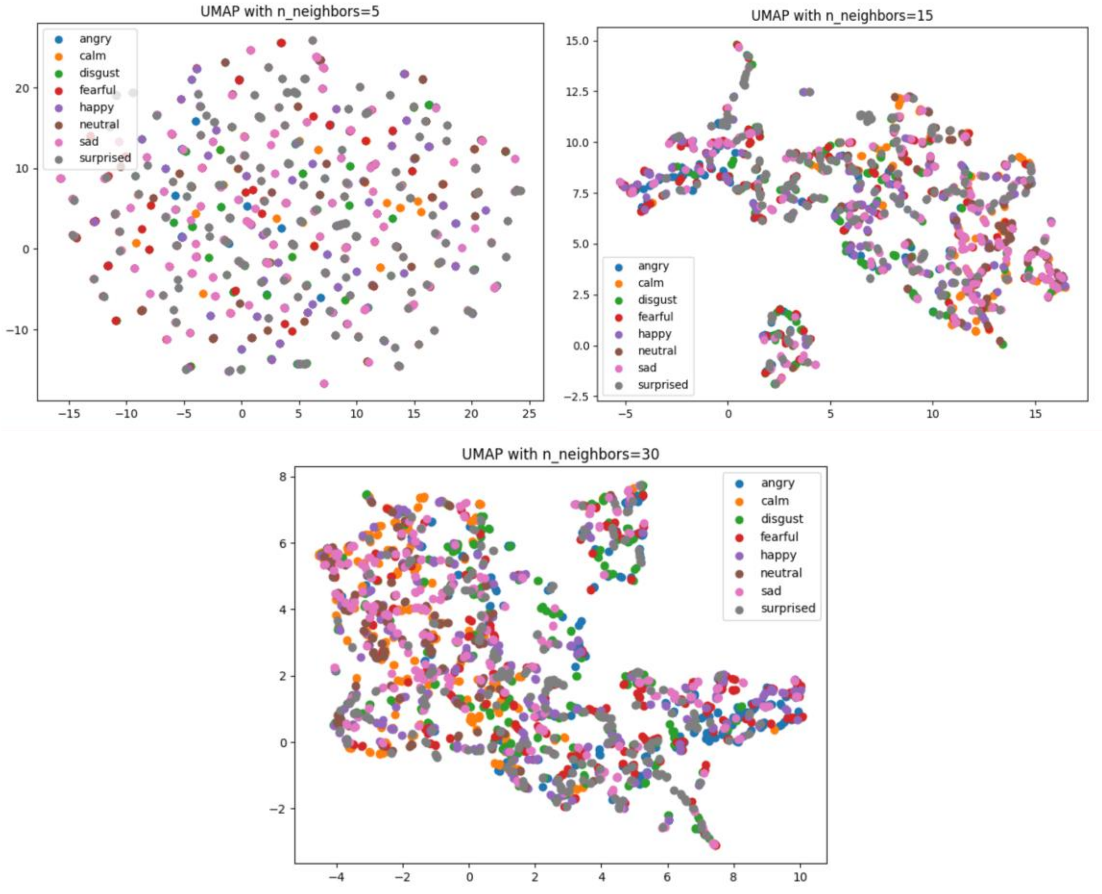
Part A

1.



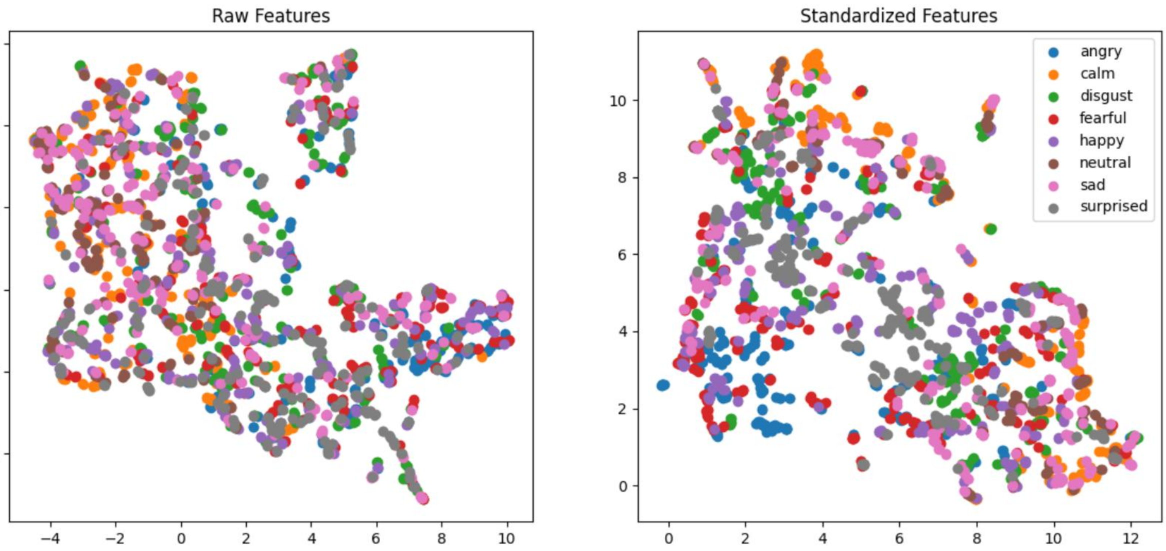
2. There are some clusters but there is a lot of overlap between clusters. Emotions like angry, happy and surprised are mixed in a region whereas emotions like sad, calm and neutral are mixed in other region. This shows that eGeMAPS features when turned into 2 dimensional features by UMAP don't have much difference and these 2 dimension features are partially or even less separable according to emotions and more dimension features are required for better boundaries.

3. As we increase the number of neighbours, we can see clusters forming and becoming better. For n=5, there is no cluster formed with every point looking at as a different boundary. As we increase n to 15 or 30, we can see clusters being formed but these 2 dimensional features are not able to distinguish between different emotions and leading to clusters formed but consisting of different emotions points.



Part B

1.



For raw features, the clusters are appearing compressed whereas for standardized features, the clusters are appearing more evenly distributed. Standardizes help in preventing features which have high value to dominate the features which help less values as clustering is depended on distance. Standardize makes all the values between 0 and 1, so all features are nearly of equal weightage. In this case since difference is only minor between the values, there is a small visible change but not an appreciable change.

2. Dimensionality reductions helps to project high- dimensional features to lower dimensions without any loss of relationships between samples. It is difficult or almost impossible to visualize higher dimensions so projecting them to lower dimensions helps in visualization of patterns, cluster and overlap. This helps us in getting info about our data like if the classes are separable, presence of outliers etc. So dimensionality reduction is a good way to explain the performance of machine learning model.

Task 5

Part A

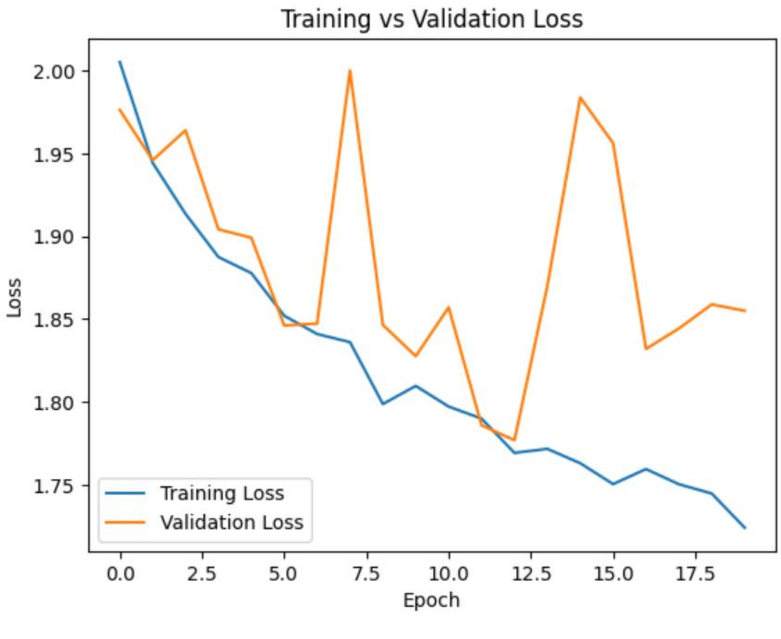
Training data= 2304 (80%)

Test data= 576 (20%)

Part B

No of parameters= 299272

Part C

1. 

2. Accuracy= 0.25
   UAR= 0.233
   The CNN accuracy and UAR were a lot lower than Task 3 models- SVM and random forest showing that CNN didn't work properly on this data. CNN was not able to properly learn features from dataset that will help to classify emotions. This could be due to size of dataset as

spectrogramas-image approach requires large data. In contrast eGeMAPS is able to work on smaller dataset.

3. Advantage of the spectrogramas-image approach compared to hand-crafted eGeMAPS features is that spectrogramas-image approach allows the model to learn acoustic patterns directly from the audio and it prevents the possibility of missing any feature done by hand-crafted eGeMAPS feature. The disadvantage is that it requires more time to train and is computationally more expensive. Since eGeMAPS are based on established features, they get trained more faster and are also easier to analyze.